

**Question 2.** Consider this code for generating random variables  $X$  and  $Y$ :

```
x = np.random.uniform()
y = np.random.geometric(p=x)
```

Derive the marginal likelihood  $\Pr_Y(y)$ , and the conditional likelihood  $\Pr_X(x | Y = y)$ .

$$X \sim U(0, 1) \quad Y | X=x \sim \text{Geo}(x)$$

$$\begin{aligned} \Pr_Y(y) &= \int_{x=0}^1 P(x, y) dx \\ &= \int_{x=0}^1 P(y|x) P(x) dx \\ &= \int_{x=0}^1 (1-x)^{y-1} x dx \\ &= \int_{z=1}^0 (z)^{y-1} (1-z) d(1-z) \quad (\text{let } x=1-z) \\ &= - \int_{z=1}^0 z^{y-1} (1-z) dz \\ &= \int_{z=0}^1 z^{y-1} - z^y dz \\ &= \left[ \frac{z^y}{y} - \frac{z^{y+1}}{y+1} \right]_0^1 \\ &= \frac{1}{y} - \frac{1}{y+1} = \frac{1}{y(y+1)} \end{aligned}$$

$$\Pr_X(x | Y=y) = \frac{P(x, y)}{P(y)} = \frac{P(y|x) P(x)}{P(y)}$$

$$= (1-x)^{y-1} x \cdot 1 \cdot y \cdot (y+1)$$

**Question 3.** I sample  $x_1, \dots, x_n$  from  $\text{Uniform}[0, \theta]$ . The parameter  $\theta$  is unknown, and I shall use  $\Theta \sim \text{Pareto}(b_0, \alpha_0)$  as my prior, where  $b_0 > 0$  and  $\alpha_0 > 1$  are known. This has the cumulative distribution function

$$\mathbb{P}(\Theta \leq \theta) = \begin{cases} 1 - (b_0/\theta)^{\alpha_0} & \text{if } \theta \geq b_0, \\ 0 & \text{if } \theta < b_0. \end{cases}$$

- Calculate the prior likelihood for  $\Theta$ .
- Show that the posterior distribution of  $(\Theta | x_1, \dots, x_n)$  is Pareto, and find its parameters.
- Find a 95% posterior confidence interval for  $\Theta$ .
- Find a different 95% posterior confidence interval. Which is better? Why?

(a) if  $\theta \geq b_0$

$$p_{\Theta}(\theta) = \frac{d \mathbb{P}(\Theta \leq \theta)}{d\theta} = \frac{d [1 - (b_0/\theta)^{\alpha_0}]}{d\theta}$$

$$= - \left( \frac{b_0^{\alpha_0}}{\theta^{\alpha_0}} \right)'$$

$$= \alpha_0 \frac{b_0^{\alpha_0}}{\theta^{\alpha_0+1}}$$

if  $\theta < b_0$ ,  $p_{\Theta}(\theta) = 0$ .

Hence,  $p_{\Theta}(\theta) = \frac{\alpha_0 b_0^{\alpha_0}}{\theta^{\alpha_0+1}} \mathbb{1}_{\theta \geq b_0}$ .

(b)  $p_{\Theta}(\theta | x_1, \dots, x_n) = \frac{p(\theta, x_1, \dots, x_n)}{p(x_1, \dots, x_n)} \propto p(x_1, \dots, x_n | \theta) p(\theta)$

$$= p(x_1 | \theta) p(x_2 | \theta) \cdots p(x_n | \theta) p(\theta)$$

$$= \frac{1}{\theta} \mathbb{1}_{x_1 \leq \theta} \frac{1}{\theta} \mathbb{1}_{x_2 \leq \theta} \cdots \frac{1}{\theta} \mathbb{1}_{x_n \leq \theta} \frac{\alpha_0 b_0^{\alpha_0}}{\theta^{\alpha_0+1}} \mathbb{1}_{\theta \geq b_0}$$

$$= \frac{\alpha_0 b_0^{\alpha_0}}{\theta^{\alpha_0+n+1}} \mathbb{1}_{\theta \geq \max\{b_0, \max_i x_i\}}$$

which is the posterior of a Pareto dist. with param

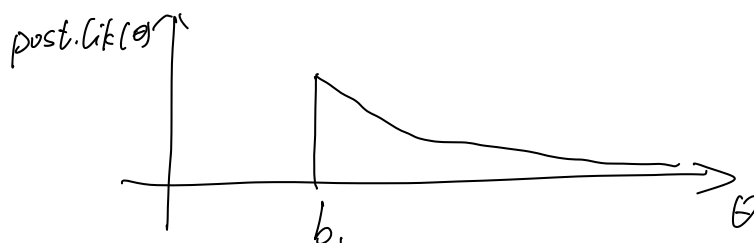
$\alpha_1 = \alpha_0 + n$  and  $b_1 = \max\{b_0, \max_i x_i\}$ .

(c) & (d)

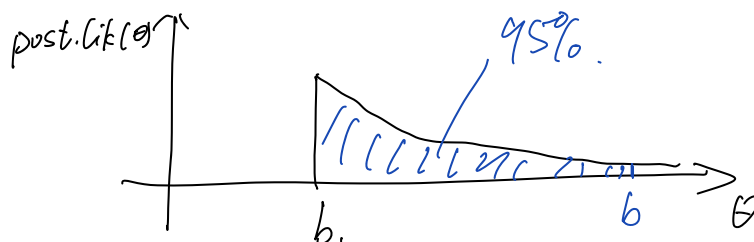
We want to find an interval  $[a, b]$  s.t.

$$P(\theta \in [a, b] | x_1, \dots, x_n) = 0.95.$$

sketch the Posterior:



first choice



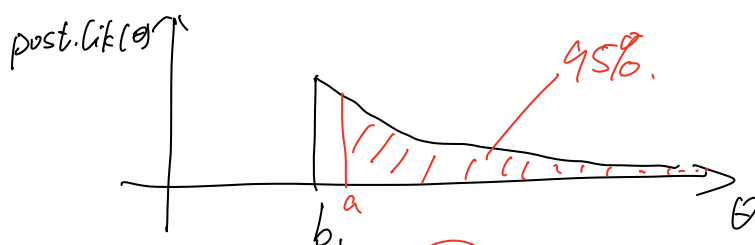
$P(\theta \in [b_1, b]) = 0.95$ , recall the cdf in Question,

$$0.95 = P(\theta \leq b) = 1 - (b_1/b)^\alpha,$$

$$\Rightarrow b = \frac{b_1}{\alpha \sqrt[0.05]{0.05}} = b_1 / (0.05)^{1/\alpha}$$

same!

another choice:



$$P(\theta \in [a, \infty)) = 0.95 \Rightarrow P(\theta \leq a) = 0.05$$

$$\Rightarrow a = b_1 / 0.95^{1/\alpha}$$

first is better: "better" is "narrowest interval"

"bags the points with the highest likelihood".

**Question 4.** I have a collection of numbers  $x_1, \dots, x_n$  which I take to be independent samples from the  $\text{Normal}(\mu, \sigma_0^2)$  distribution. Here  $\sigma_0$  is known, and  $\mu$  is unknown. Using the prior distribution  $M \sim \text{Normal}(\mu_0, \rho_0^2)$  for  $\mu$ , show that the posterior density is

$$\Pr_M(\mu | x_1, \dots, x_n) = \kappa e^{-(\mu-c)^2/2\tau^2}$$

where  $\kappa$  is a normalizing constant, and where you should find formulae for  $c$  and  $\tau$  in terms of  $\sigma_0$ ,  $\mu_0$ , and  $\rho_0$ , and the  $x_i$ . Hence deduce that the posterior distribution is  $\text{Normal}(c, \tau^2)$ . [Note: 'M' is the upper-case form of the Greek letter 'mu'.]

$$M \sim \text{Normal}(\mu_0, \rho_0^2) \Rightarrow \Pr_M(\mu) = \text{const} \cdot \exp(-(\mu-\mu_0)^2/2\rho_0^2)$$

$$\begin{aligned} X \sim \text{Normal}(\mu, \sigma_0^2) \Rightarrow \Pr(x_1 \dots x_n | \mu) &= \text{const} \cdot \prod_{i=1}^n e^{-(x_i-\mu)^2/2\sigma_0^2} \\ &= \text{const} \cdot \exp(-\sum_i (x_i-\mu)^2/2\sigma_0^2) \end{aligned}$$

Then, the posterior is

$$\begin{aligned} \Pr_M(\mu | x_1 \dots x_n) &= \text{const} \cdot \Pr_M(\mu) \cdot \Pr(x_1 \dots x_n | \mu) \\ &= \text{const} \cdot \exp\left[-\frac{(\mu-\mu_0)^2}{2\rho_0^2} - \frac{\sum_i (x_i-\mu)^2}{2\sigma_0^2}\right] \\ &= \text{const} \cdot \exp\left[-\frac{\mu^2 - 2\mu\mu_0 + \mu_0^2}{2\rho_0^2} - \frac{\sum_i x_i^2 - 2\mu\sum_i x_i + \mu^2}{2\sigma_0^2}\right] \\ &= \text{const} \cdot \exp\left[-\mu^2 \left(\frac{1}{2\rho_0^2} + \frac{n}{2\sigma_0^2}\right) + 2\mu \left(\frac{\mu_0}{2\rho_0^2} + \frac{\sum_i x_i}{2\sigma_0^2}\right)\right] \\ &= \text{const} \cdot \exp\left[-\frac{1}{2\tau^2} \left\{ \mu^2 - 2\mu\tau^2 \left(\frac{\mu_0}{\rho_0^2} + \frac{\sum_i x_i}{\sigma_0^2}\right) \right\}\right] \\ &= \text{const} \cdot \exp\left[-\frac{1}{2\tau^2} \left\{ \mu - \tau^2 \left(\frac{\mu_0}{\rho_0^2} + \frac{\sum_i x_i}{\sigma_0^2}\right) \right\}^2\right] \\ &= \text{const} \cdot \exp\left[-\frac{(\mu-c)^2}{2\tau^2}\right] \quad \text{where } c = \tau^2 \left(\frac{\mu_0}{\rho_0^2} + \frac{\sum_i x_i}{\sigma_0^2}\right) \end{aligned}$$

The posterior distribution for  $\mu$  is  $\text{Normal}(c, \tau^2)$  where

$$\tau = \frac{1}{\sqrt{\frac{1}{\rho_0^2} + \frac{n}{\sigma_0^2}}} \quad c = \frac{\frac{\mu_0}{\rho_0^2} + \frac{\sum_i x_i}{\sigma_0^2}}{\frac{1}{\rho_0^2} + \frac{n}{\sigma_0^2}}$$

**Question 5 (Leaky priors).** I repeatedly attempt a task, and each time I attempt it I succeed with probability  $\theta$  and fail with probability  $1 - \theta$ . The parameter  $\theta$  is unknown, so I model it as a random variable  $\Theta$ . Ever the optimist, my prior for  $\Theta$  is heavily biased in favour of large values for  $\theta$ :

$$\Pr_{\Theta}(\theta) = \varepsilon 1_{\theta \leq 1/2} + (2 - \varepsilon) 1_{\theta > 1/2}$$

for some known small value  $\varepsilon > 0$ ; this implies  $\mathbb{P}(\Theta \leq 1/2) = \varepsilon/2$ .

But I experience an unbroken run of  $n$  failures. How big does  $n$  need to be, for me to concede there's a 50% posterior probability that  $\Theta \leq 1/2$ ? How big would it need to be, if  $\varepsilon = 0$ ?

Firstly, we need the posterior probability.

$$\begin{aligned} P(\Theta | x_1 \dots x_n) &= \text{const} \cdot P(x_1 \dots x_n | \Theta) P(\Theta) \\ &= \text{const} \cdot (1 - \theta)^n (\varepsilon 1_{\theta \leq \frac{1}{2}} + (2 - \varepsilon) 1_{\theta > \frac{1}{2}}) \\ &= \begin{cases} C \varepsilon (1 - \theta)^n & \text{if } \theta \leq \frac{1}{2} \\ C (2 - \varepsilon) (1 - \theta)^n & \text{if } \theta > \frac{1}{2} \end{cases} \end{aligned}$$

We could get  $C$  by  $\int P(\Theta | x_1 \dots x_n) d\theta = 1$ . But no need for this question.

We want  $P(\Theta \leq \frac{1}{2}) = 50\%$ , that means

$$P(\Theta \leq \frac{1}{2}) = P(\Theta > \frac{1}{2})$$

$$\int_{\theta=0}^{\frac{1}{2}} C \varepsilon (1 - \theta)^n d\theta = \int_{\theta=\frac{1}{2}}^1 C (2 - \varepsilon) (1 - \theta)^n d\theta$$

$$\frac{C \varepsilon}{n+1} \left(1 - \left(\frac{1}{2}\right)^{n+1}\right) = \frac{C (2 - \varepsilon)}{n+1} \left(\frac{1}{2}\right)^{n+1}$$

$$\varepsilon \left(1 - \left(\frac{1}{2}\right)^{n+1}\right) = (2 - \varepsilon) \left(\frac{1}{2}\right)^{n+1}$$

$$\varepsilon - \varepsilon \left(\frac{1}{2}\right)^{n+1} = \left(\frac{1}{2}\right)^n - \varepsilon \left(\frac{1}{2}\right)^{n+1}$$

$$n = -\log_2 \varepsilon.$$

If  $\varepsilon = 0$  then  $P(\Theta \leq \frac{1}{2}) = 0$  whatever the data.

**Question 6.** I have a collection of numbers

[4.3, 2.8, 3.9, 4.1, 9, 4.5, 3.3]

which look like they mostly come from a Gaussian distribution, but with the occasional outlier. Model the data as

$$X \text{ is } \begin{cases} \text{Normal}(\mu, 0.5^2) & \text{with probability 99\%} \\ \text{Cauchy} & \text{with probability 1\%.} \end{cases}$$

Use a  $\text{Normal}(0, 5^2)$  prior distribution for  $\mu$ . Give pseudocode to plot the posterior distribution. [Note. The Cauchy random variable occasionally generates wildly huge values. The library function `scipy.stats.cauchy.pdf(x)` computes its pdf.]

Recall the lecture (slide 10)

Computational Bayes

1. Take a sample  $\theta_1 \dots \theta_n$  from  $\Theta$ .
2. Compute weights  $w_i = \text{Pr}_x(x | \theta = \theta_i)$   
and rescale sum to 1.
3. Reason about the posterior via the approximation  $\mathbb{P}(\theta \in A | X=x) \approx \sum_i w_i \mathbb{I}_{\theta_i \in A}$ .

All we need to calc by hand is  $\text{Pr}_x(x | \mu)$ :

$$\begin{aligned} \text{cdf}(x) &= \mathbb{P}(X \leq x | \mu) \\ &= \mathbb{P}(X \leq x | \mu, \text{non-outlier}) \mathbb{P}(\text{non-outlier}) + \\ &\quad \mathbb{P}(X \leq x | \mu, \text{outlier}) \mathbb{P}(\text{outlier}) \\ &= 0.99 \text{cdf}_{\text{Normal}(\mu, 0.5)}(x) + 0.01 \text{cdf}_{\text{Cauchy}}(x) \end{aligned}$$

then we have the code

```
1 x = [4.3, 2.8, 3.9, 4.1, 9, 4.5, 3.3]
2 def prx(x, μ):
3     p1 = scipy.stats.norm.pdf(x, loc=μ, scale=0.5)
4     p2 = scipy.stats.cauchy.pdf(x)
5     return np.prod(0.99*p1 + 0.01*p2)
6
7 μsamp = np.random.normal(loc=0, scale=5, size=3000)
8 w = np.array([prx(x, μ) for μ in μsamp])
9 w = w / np.sum(w)
10
11 plt.hist(μsamp, weights=w, density=True, bins=np.linspace(2, 7, 60))
12 plt.show()
```

**Question 7.** In lecture notes section 2.6 we investigated a dataset of police stop-and-search actions. Let the outcome for record  $i$  be  $y_i \in \{0, 1\}$ , where 1 denotes that the police found something and 0 denotes that they found nothing. Consider the probability model  $Y_i \sim \text{Binom}(1, \beta_{\text{eth}_i})$  where  $\text{eth}_i$  is the recorded ethnicity for the individual involved in record  $i$ , and where the parameters  $\beta_{\text{As}}, \beta_{\text{Blk}}, \beta_{\text{Mix}}, \beta_{\text{Oth}}, \beta_{\text{Wh}}$  are unknown. As a prior distribution, suppose that the five  $\beta$  parameters are all independent  $\text{Beta}(1/2, 1/2)$  random variables.

- (a) Write down the joint prior density for  $(\beta_{\text{As}}, \beta_{\text{Blk}}, \beta_{\text{Mix}}, \beta_{\text{Oth}}, \beta_{\text{Wh}})$ .  
 (b) Find the joint posterior distribution of  $(\beta_{\text{As}}, \beta_{\text{Blk}}, \beta_{\text{Mix}}, \beta_{\text{Oth}}, \beta_{\text{Wh}})$  given the  $y$  data.

(a) Recall the density of Beta distribution

$$\text{Pr}_{\text{Beta}(\alpha, \beta)}(x) = \text{const} \cdot x^{\alpha-1} (1-x)^{\beta-1}.$$

then the joint prior distribution is

$$\text{Pr}(\beta_{\text{As}}, \beta_{\text{Blk}}, \beta_{\text{Mix}}, \beta_{\text{Oth}}, \beta_{\text{Wh}}) = \text{const} \cdot \prod_e \beta_e^{-\frac{1}{2}} (1-\beta_e)^{-\frac{1}{2}}.$$

where  $e \in \{\text{As}, \text{Blk}, \text{Mix}, \text{Oth}, \text{Wh}\}$

(b) we have prior, we want posterior, so let's calc the likelihood.

$$Y_i \sim \text{Binom}(1, \beta_{\text{eth}_i})$$

$$\text{Pr}(Y_i = y) = \beta_{\text{eth}_i}^y (1 - \beta_{\text{eth}_i})^{(1-y)}$$

$$\text{then } \text{Pr}(y_1, \dots, y_n | \vec{\beta}) = \text{const} \cdot \prod_{i=1}^n \beta_{\text{eth}_i}^{y_i} (1 - \beta_{\text{eth}_i})^{(1-y_i)}$$

$$= \text{const} \cdot \prod_e \beta_e^{s_e} (1 - \beta_e)^{n_e - s_e}$$

where  $n_e$ :  $\# \text{eth}_i = e$ .

$s_e$ :  $\# y_i = 1$  and  $\text{eth}_i = e$

the posterior

$$\text{Pr}(\vec{\beta} | y_1, \dots, y_n) = \text{const} \cdot \text{Pr}(y_1, \dots, y_n | \vec{\beta}) \cdot \text{Pr}(\beta_{\text{As}}, \beta_{\text{Blk}}, \beta_{\text{Mix}}, \beta_{\text{Oth}}, \beta_{\text{Wh}})$$

$$= \text{const} \cdot \prod_e \beta_e^{-\frac{1}{2}} (1 - \beta_e)^{-\frac{1}{2}} \beta_e^{s_e} (1 - \beta_e)^{n_e - s_e}$$

$$= \text{const} \cdot \prod_e \beta_e^{s_e - \frac{1}{2}} (1 - \beta_e)^{n_e - s_e - \frac{1}{2}}$$

which is the joint density of 5 indep. Beta dist.s